

6.4.4. Nonideal Practical Sampling Analysis

Up to this point, we have always used an ideal impulse sampling pulse train to exactly retrieve the value $g(kT_s)$ precisely at $t = kT_s$ for uniform signal sampling. This is not possible to realize in practice, and an alternate, more practical method of sampling needs to be considered. Analyses of these methods are important to help us understand the errors that occur during practical analog-to-digital (A/D) conversion, and the effects they have on signal reconstruction.

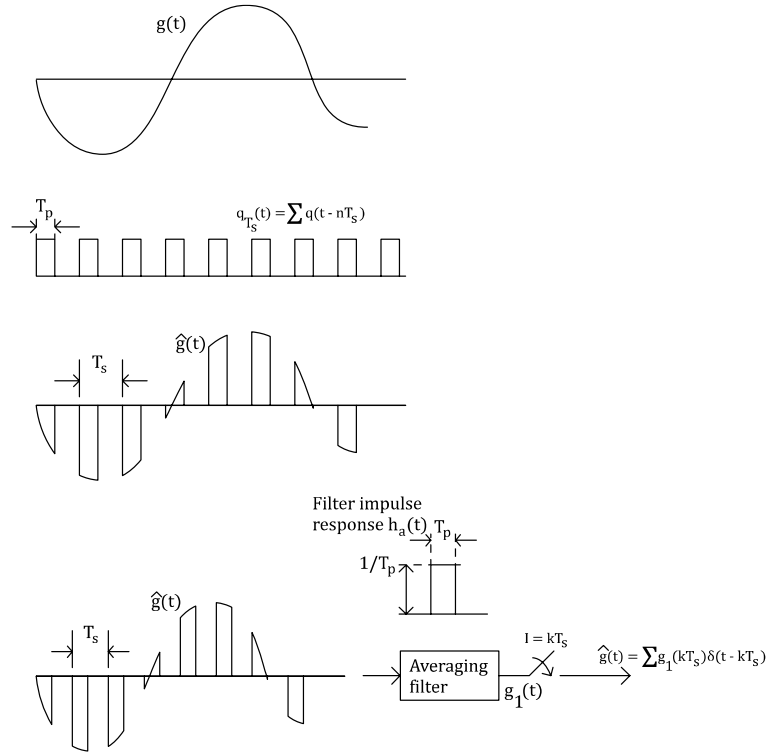


Figure 6.14. Illustration of practical sampling.

Practical samplers do not take samples at an instantaneous moment in time, but rather for a finite duration T_p centered about kT_s . Therefore, these samplers take a snapshot of duration T_p every T_s seconds. This is similar to a continuous frame capture in cameras, where still-photographs are taken in equal time intervals. The cameras generate a still-picture by taking the average of the entire picture scene of T_p seconds, similar to how practical samplers would generate a sample at $t = T_s$ by averaging the signal set over the window T_p , described by:

$$g_1(kT_s) = \frac{1}{T_p} \int_{-T_p/2}^{T_p/2} g(kT_s + t) dt \quad (6.12)$$

In turn, practical samplers will form this sampled signal:

$$\tilde{g}(t) = \sum g_1(kT_s) \delta(t - kT_s) \quad (6.13)$$

Thus, there will be induced errors due to the way the samples were taken.

6.5 Pulse-Amplitude Modulation (PAM)

Since we now learned the practical sampling method, we can now define the simplest of analog pulse modulation: *pulse amplitude modulation* (PAM). In PAM, evenly spaced pulses have their amplitudes varied proportionally to the corresponding continuous message signal sample. These pulses can be rectangular, or some other shape. PAM is

similar to zero-order hold signal reconstruction from signal samples; but in PAM, rectangular pulse width is much shorter than the sampling period.

Figure 6.15 shows an example waveform of a PAM signal, where the dashed curve is the waveform of a message signal $m(t)$, and the amplitude-modulated rectangular pulse sequence shown in solid lines represents the corresponding PAM signal $s(t)$. Two operations are involved with PAM signal generation:

- Instantaneous sampling of $m(t)$ every T_s seconds, with the chosen sampling rate $f_s = 1/T_s$ in accordance with the sampling theorem
- Lengthening the duration of every sample of a constant duration T

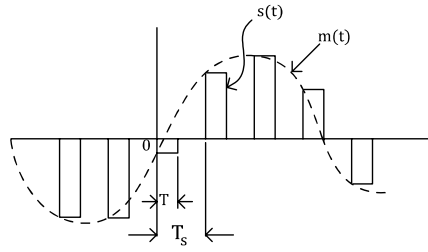


Figure 6.15 Flat-top samples, representing an analog signal.

The two aforementioned PAM signal generation operations are each known respectively as "sample, and hold" in digital circuit technology. A reason for intentionally lengthening the duration of each sample is to limit channel bandwidth use, as bandwidth is inversely proportional to sample duration. Obviously, we should be careful into choosing how large/small we make the sample duration T , which we will show in the following analysis.

Let $s(t)$ denote the pulse train generated in the manner of Figure 6.15 We can then express the PAM signal as:

$$s(t) = \sum_{n=-\infty}^{\infty} m(nT_s)h(t - nT_s) \quad (6.14)$$

where T_s is the sampling period, $m(nT_s)$ is the sampled value of $m(t)$ at $t = nT_s$, and $h(t)$ is a standard rectangular pulse of unit amplitude and duration T , defined as (also see Figure 6.16(a)):

$$h(t) = \begin{cases} 1 & 0 < t < T \\ 0 & \text{otherwise} \end{cases} \quad (6.15)$$

The instantaneously sampled impulse train $m(t)$ by definition, is given by:

$$m_\delta(t) = \sum_{n=-\infty}^{\infty} m(nT_s)\delta(t - nT_s) \quad (6.16)$$

where $\delta(t - nT_s)$ is a time-shifted delta function. Convoluting $m_\delta(t)$ with the pulse $h(t)$ will therefore give us:

$$\begin{aligned} m_\delta(t) * h(t) &= \int_{-\infty}^{\infty} m_\delta(\tau)h(t - \tau)d\tau \\ &= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} m(nT_s)\delta(\tau - nT_s)h(t - \tau)d\tau \\ &= \sum_{n=-\infty}^{\infty} m(nT_s) \int_{-\infty}^{\infty} \delta(\tau - nT_s)h(t - \tau)d\tau \end{aligned} \quad (6.17)$$

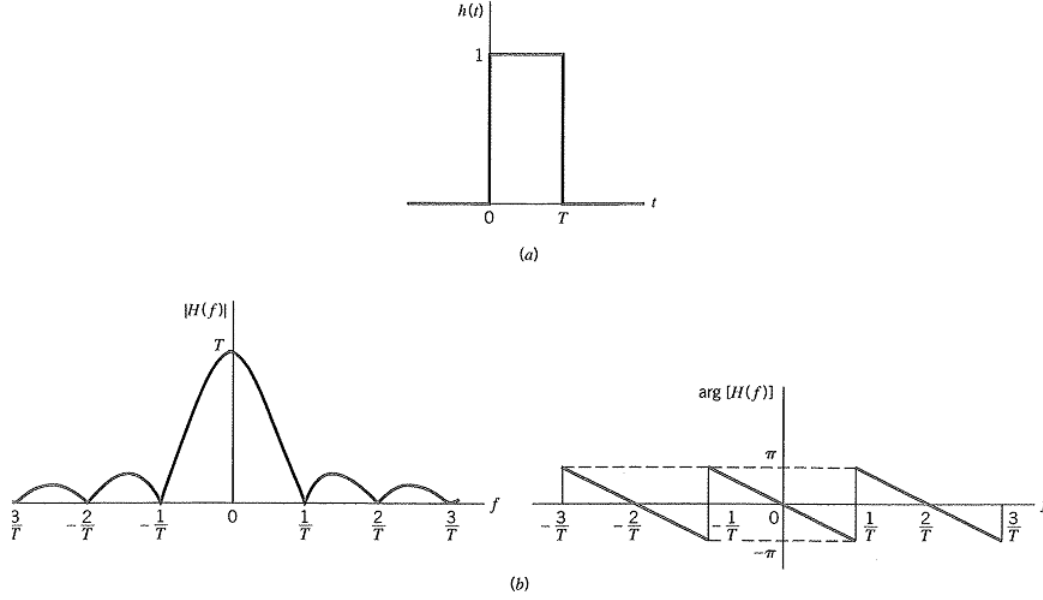


Figure 6.16 (a) Rectangular pulse $h(t)$. (b) Spectrum $H(f)$, made up of the magnitude $|H(f)|$, and phase $\arg\{H(f)\}$.

We obtain the following by using the delta function's shifting property:

$$m_\delta(t) * h(t) = \sum_{n=-\infty}^{\infty} m(nT_s) h(t - nT_s) \quad (6.18)$$

Equations 5.15, and 5.19 show how the PAM signal $s(t)$ is mathematically equivalent to the convolution of the instantaneously sampled $m(t)$, $m_\delta(t)$, and the pulse $h(t)$, as shown by

$$s(t) = m_\delta(t) * h(t) \quad (6.19)$$

If we take the Fourier transform of Equation 5.20, recognizing that the convolution of two time signals results in the Fourier transform of the product of the two signal's Fourier transform:

$$S(\omega) = M_\delta(\omega) H(\omega) \quad (6.20)$$

Since

$$M_\delta(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} M(\omega - k\omega_s) \quad (6.21)$$

where f_s is the sampling rate. Substituting equation 5.22 into 5.21 yields

$$S(\omega) = H(\omega) \cdot \frac{1}{T_s} \sum_{k=-\infty}^{\infty} M(\omega - k\omega_s) \quad (6.22)$$

How would we recover the original message signal $m(t)$ given the PAM signal $s(t)$ with the Fourier transform of $S(\omega)$? As a first step, we can pass $s(t)$ through a lowpass filter with a bandwidth of $\omega_s / 2$. The output will now have a spectrum $M(\omega)H(\omega)$, which equivalent to passing the original message signal $m(t)$ through a low-pass filter of frequency response $H(\omega)$.

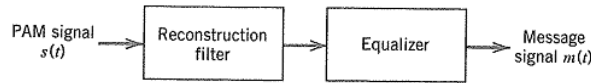


Figure 6.17 System for recovering message signal $m(t)$ from PAM signal $s(t)$.

Equation 5.16 shows that the Fourier transform of the rectangular pulse $h(t)$ (plotted in Figure 5.17(b)) is given by

$$H(\omega) = T \operatorname{sinc}(\omega T / 2\pi) \exp(-j\omega T / 2) \quad (6.23)$$

We can now see that by using flat-top samples to generate a PAM signal, an amplitude distortion and a delay of $T/2$ is introduced. This is known as the aperture effect, the distortion caused by the use of PAM to transmit analog signals.

This distortion can be corrected by another filter that cascades with the low-pass reconstruction filter such that distortions caused by the reconstruction filter will be cancelled out, and these filters that fix the distortion are called *equalizers*. The magnitude response of the equalizer in this case is ideally given by:

$$\frac{1}{|H(\omega)|} = \frac{1}{T \text{sinc}(\omega T / 2\pi)} = \frac{\omega / 2}{\sin(\omega T / 2)} \quad (6.24)$$

In practice, the amount of equalization needed is usually small. For a duty cycle with $T/T_s \leq 0.1$, the amplitude distortion is under 0.5%, equalization can be omitted entirely.

PAM signal transmission imposes strict requirements on channel magnitude and phase responses due to the relatively short duration of the transmitted pulses.

6.6 Other Forms of Pulse Modulation

In a pulse modulation system we can encode the sample values of the message signal in a property of the pulse other than amplitude for signal transmission:

Pulse-width modulation (PWM), also known as pulse-duration modulation, message signal samples are used to vary the duration of the individual pulses in the carrier.

Pulse-position modulation (PPM), has its pulse position placed relative to its un-modulated time of occurrence varied in accordance with the message signal.

The two aforementioned forms of pulse modulation are shown in Figure 5.19 for a sinusoidal message signal.

PWM expends lots of power for long pulses and does not give more information. If this unused power is subtracted from PWM so that only time transitions remain, PPM is obtained. PPM is a more efficient form of pulse modulation than PWM.

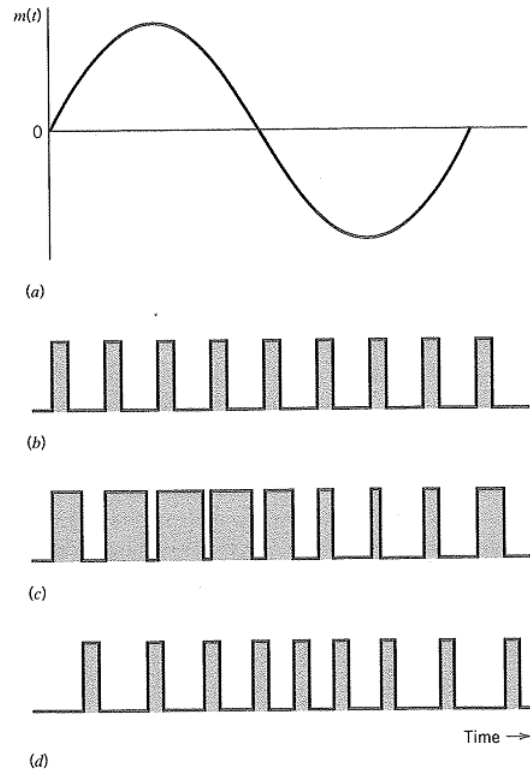


Figure 6.18 Illustrating two different forms of pulse-time modulation for the case of a sinusoidal modulating wave. (a) Modulating wave. (b) Pulse carrier. (c) PWM wave. (d) PPM wave.

6.7 Pulse Code Modulation (PCM)

Now that we learned about PAM, we can introduce digital transmission. PCM (Pulse-code modulation) is the most basic of digital pulse modulation. In PCM, the message signal is represented discretely in both time and amplitude. Representing a signal in this form allows it to be transmitted as a sequence of coded binary pulses.

Sampling, and *quantization* are the two fundamental processes in binary PCM wave generation. Sampling, as described before, is the process that generates the message signal's discrete-time representation, and we have to follow the sampling theorem described previously. Quantization is the process that generates the message signal's discrete-amplitude representation, which will shortly be described.

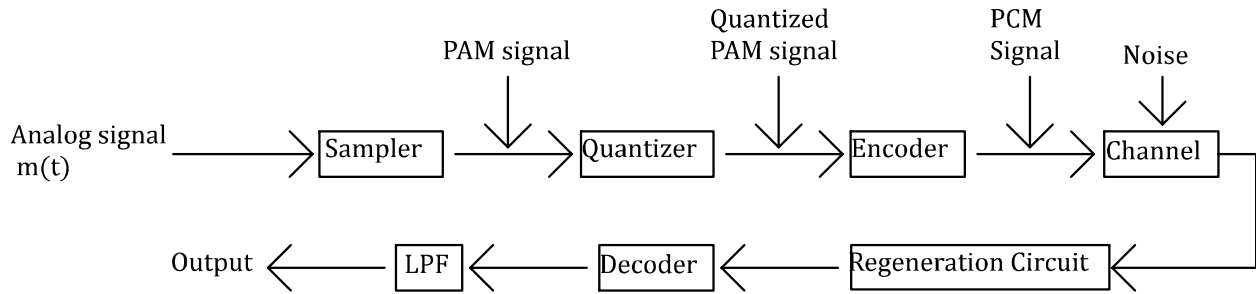


Figure 6.19 A PCM system.

6.7.1. Quantization

An analog signal can have an amplitude that can have any value over a continuous range, which means that an amplitude in an analog signal can be an infinite number of values. Digital signals can only take on a finite number of values. Analog signals can be converted into digital signals by *sampling* and *quantization*, rounding off values to the closest permitted values (quantized levels), and Figure 6.20 shows an example analog signal getting converted into a digital signal.

One such type of digital signal takes the range of an analog signal's amplitude $(-m_p, m_p)$, and partitions into L subintervals, with the spacing between each of them being $\Delta v = 2m_p/L$. Every sample amplitude is approximated by the midpoint value of the subinterval where the sample falls. Each quantized signal is then approximated to one of the L numbers. This signal is therefore digitized, and this signal is known as an L -ary digital signal.

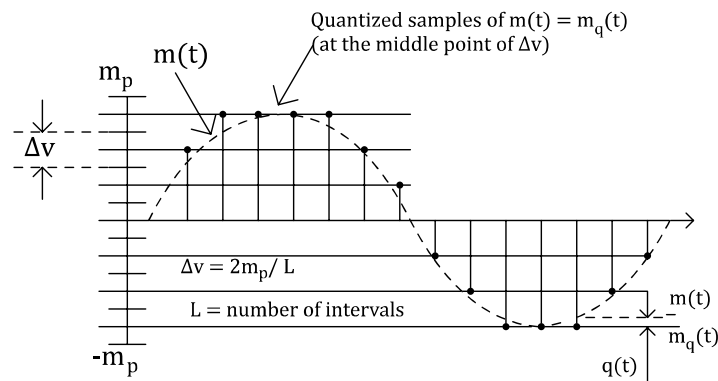


Figure 6.20 Quantization of a sampled analog signal.

A quantizer can be uniform or non-uniform, and midtread or midrise. A uniform quantizer takes uniformly spaced out levels; if it isn't, then it is a non-uniform quantizer. A midtread quantizer has its origin in the middle of a tread in a staircase-like graph (Figure 6.21 (a)), whereas a midrise quantizer has its origin in the middle of a vertical segment of

the staircase-like graph (Figure 6.21(b)). Both midtread and midrise uniform quantizers in Figure 6.21 are symmetric about the origin.

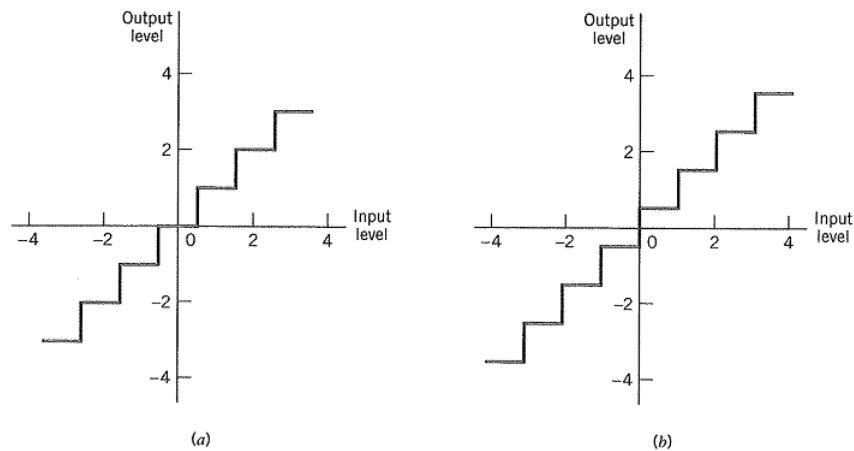


Figure 6.21. Two types of quantization: (a) mid tread and (b) midrise.

6.7.2. Quantization Noise

Quantization noise is an error that is defined as the difference between the input and the output signal. Figure 6.22 illustrates the quantization noise as a function of time, assuming the use of a uniform midtread quantizer. Quantization noise will be further discussed later on.

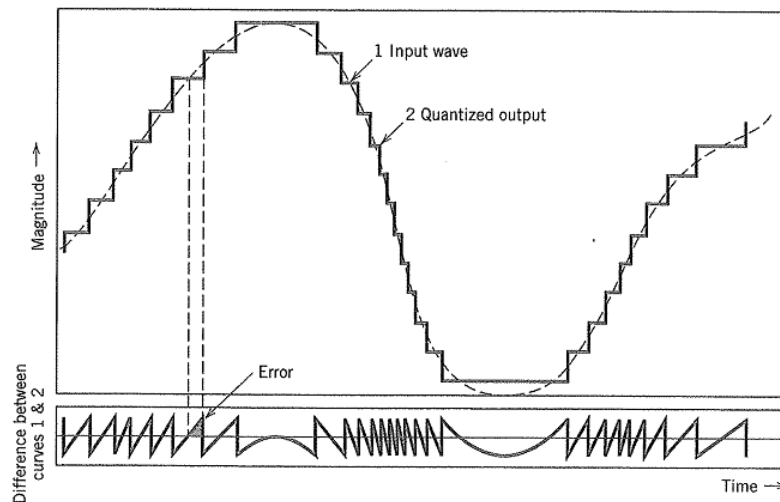


Figure 6.22 Illustration of the quantization process. (Adapted from Bennett, 1948, with permission of AT&T.).

Binary digital signal (signal that can take on just two values) is desirable practically because it is simple, economic, and easy to use. L-ary to binary signal conversion can be done by pulse encoding.

6.7.3. Encoding

After a message signal gets sampled and quantized, it won't be in a form best suited for transmission. To make the transmitted signal robust to noise and interference, we would need to transform this discrete sample set to an appropriate form of signal, and this is called the *encoding process*. The encoded values are usually combinations of *binary digits*, which can be transmitted by assigning a distinct pulse shape to each of the two bits. Figure 6.23 shows

a binary signal constructed by having each sample transmitted by a group of three binary pulses, with binary 0 being no pulse, and binary 1 being a positive pulse.

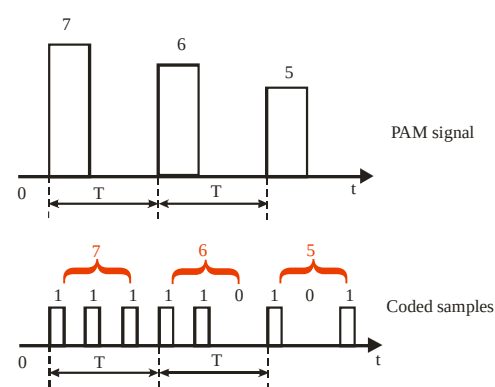


Figure 6.23 Signal quantization and binary code assignment.

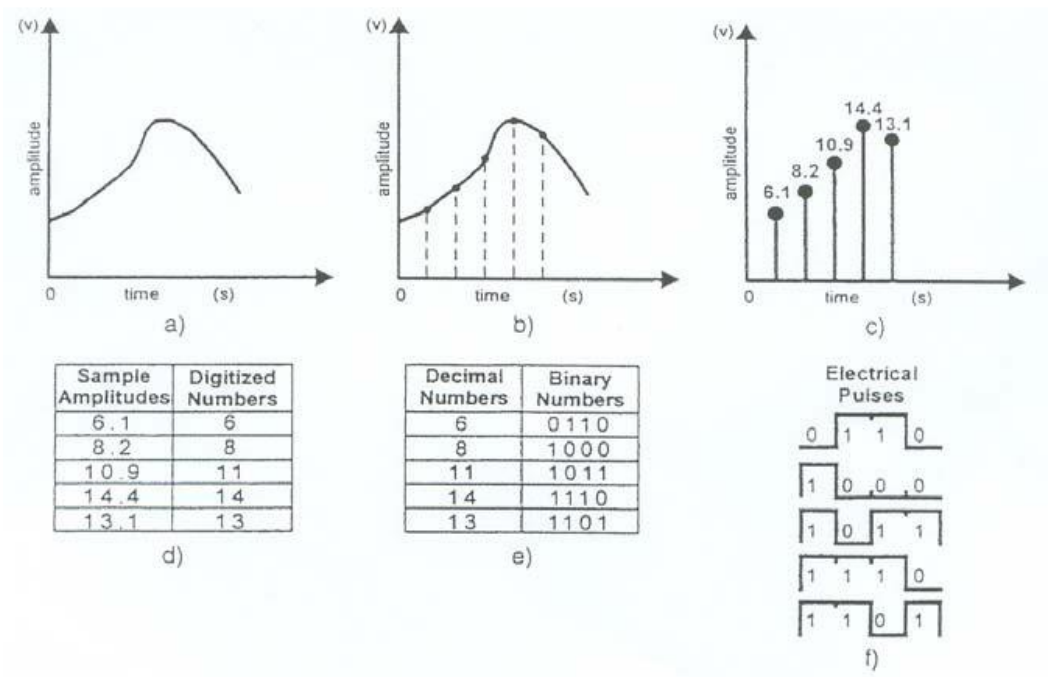


Figure 6.24 Quantization and encoding in PCM: a) Original analog signal; b) sampling procedure; c) obtained samples; d) quantization; e) binary coding; f) electrical signal transmitted in digital form.

Typically, 8 bits is called a byte.

6.7.4. Digital encoding of speech and music

The audio signal bandwidth is typically around 4 kHz. The intelligibility of speech however, is not affected if components higher than 3.4 kHz are suppressed. This is why telephone communication filters components higher than 3.4 kHz with a low-pass filter, where communication is intelligibility rather than high fidelity. The output signal is then sampled at 8 kHz (8000 samples per second), which is intentionally higher than the Nyquist rate of 6.8 kHz because then it would be realizable to implement practical filters for signal reconstruction. Every sample is then quantized into 1 of 256 levels ($L = 256$), which requires eight binary pulses (8 bits, or 1 byte) to encode each sample

($2^8 = 256$). A telephone signal therefore transmits $8 \times 8000 = 64000$ binary pulses per second or 64 kbit/s. This is also known as a *DS0 signal* in the literature.

A newer application of the PCM is the compact disc (CD). CD music is a high-fidelity, and requires that the audio signal bandwidth be 20 kHz. The sampling rate is 44.1 kHz, even though the Nyquist rate is much smaller at 20 kHz, due to the aforementioned reason of filter implementation practicality. The number of quantization levels is very high also, at $L = 65,536$ (16 bits, $2^{16} = 65,536$) to reduce the quantizing error. The binary-coded samples (1.5 MBit/s) are then recorded on the CD.

6.8 Advantages of Digital Communication

There are a number of advantages that digital communication offers over analog communication, some of which are listed below.

1. The main advantage that digital communications offers over analog communication is viability of regenerative repeaters. Both digital and analog signals get inhibited by noise, interference, and distortion. Analog signals also get weaker as it travels further down a channel. Analog repeaters amplify the message signal, but the noise and distortions that are already inside the message signal would be amplified too, ultimately overwhelming the original message signal. In digital communications, a repeater is set up at distances short enough to detect the signal pulses before noise and distortion have a chance to sufficiently ruin the signal. These pulses are detected at each repeater, and new, clean pulses are transmitted to the next repeater station, and this process continues. Pulses can be detected reliably and correctly if the repeaters are close enough such that the noise and distortion are within acceptable limits. Therefore, digital messages are more reliable over longer distances. The largest errors that exist in PCM comes from quantizing, which can be reduced as much as desired by increasing the number of quantized levels, with the price being the increase in channel bandwidth.
2. The flexibility of digital hardware implementation permits the use of digital switching, large-scale integrated circuits, and microprocessors.
3. It is possible to encode digital signals to yield low error rates, high fidelity, and higher security.
4. Digital signal multiplexing is easier and more efficient.
5. Digital communication is inherently more effective than analog in exchanging SNR for bandwidth.
6. Storing digital signals is relatively easy and cheap. Digital signal storage can also search and select information from an electronic database.
7. Digital message reproduction is very reliable with zero deterioration, contrasting to analog messages which loses quality every successive stage of reproduction, and also has to be physically transported from one point to another which can be costly.
8. Digital hardware halves in cost and doubles in performance and storage capacity every two to three years (similar to Moore's law). This progress has not been stopped and is still continuing; as a result, digital technologies are the dominant choice in any area of communication or storage technologies.
9. Digital communication can work with noise and distortion better than analog communication as long as the noise and distortion are within limits. Analog signals will be distorted by any noise or distortion no matter how small it is.

6.9 Digital Telephony: PCM in T1 Carrier Systems

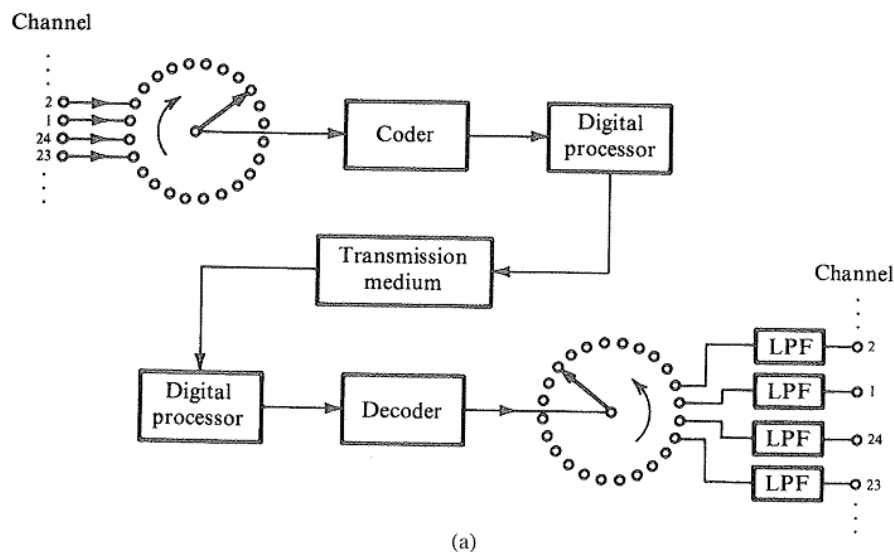
6.9.1. A Historical Note

The time between PCM's invention and implementation took a lengthy time of more than 20 years because of the non-existence of any practical switching devices. Vacuum tubes were used before the transistor was invented, and they were large, poor switches, and dissipated much heat. Therefore, systems that incorporated vacuum tubes were large, unreliable, and overheated often. The transistor's timely invention allowed PCM to be practical because it was small, consumed little power, and was almost an ideal switch.

The invention of the transistor was timely due to the high demand of the phone service, and the overload of the existing system, especially in large cities. Underground space was not easily accessible for cable installation in many cities because they were already used by other services such as gas, sewage, and hydro. Digging the streets weren't very desirable either. An attempt of using FDM to increase capacity by multiplexing multiple voice channels with amplitude modulation. These cables are designed primarily for the audible voice range of 0 to 4 kHz, and suffered from noise as a result. Cross talk between pairs of channels on the same cable was also unacceptable at high frequencies. PCM however, offers the solution even though it requires a higher bandwidth than that required for FDM signals because repeaters can be spaced closely enough on digital systems so that noisy lines won't be a big issue. Repeaters spaced approximately 6000 feet apart refresh the signal and send out a new clean pulse before it gets too noisy/distorted. This is the history of Bell System's T1 carrier system. As a result, the pair of wires that once transmitted a single 4 kHz bandwidth audio signal is used to transmit 24 TDM PCM phone signals with a 1.544 MHz bandwidth.

6.9.2. T1 Time Division Multiplexing

Figure 6.25(a) shows the T1 carrier system schematic. All 24 channels are time division multiplexed, and the sampler output is a TDM PAM signal. An encoder then takes the signal and quantizes it, then encodes it into 8 bit pulses (8 binary pulse), shown in Figure 6.25 (b). The signal is thus digitized, and sent over a channel. Repeaters are placed every 6000 feet to refresh these pulses, and are decoded (binary pulse to sample conversion) at the receiver. These samples are then demultiplexed and the audio signal is reconstructed by passing the samples through a low-pass filter in each channel.



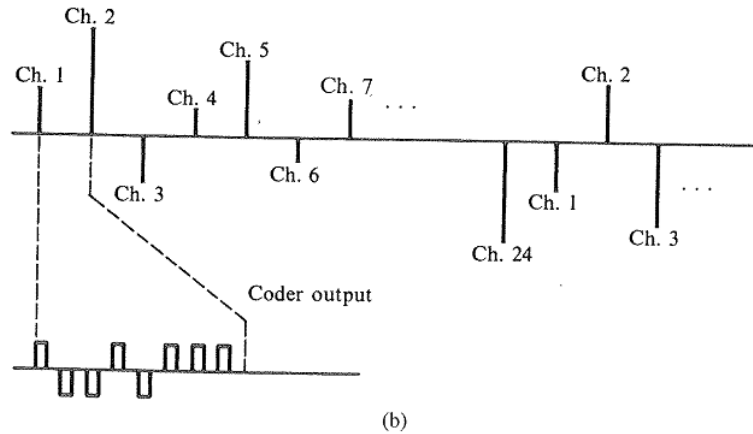


Figure 6.25. A T1 carrier system.

Commutators are electronic switching circuits (Figure 6.25). The 1.544 Mbit/s signal (also called digital signal level 1, or DS1) can be multiplexed into higher level signals DS2, DS3, and DS4 as shown in Figure 6.27. After the T1 carrier system was introduced, many variations of this system was adopted/proposed before it was standardized by the ITU-T (International Telecommunication Union) into a 30-channel PCM system with a transfer rate of 2.049 Mbit/s (T1 has a 24 channel 1.544 Mbit/s system). This standard is used worldwide with the US and Japan being the exception because they adopted T1 before the ITU-T standard was introduced. As a result, these two standards are still used in the world today.

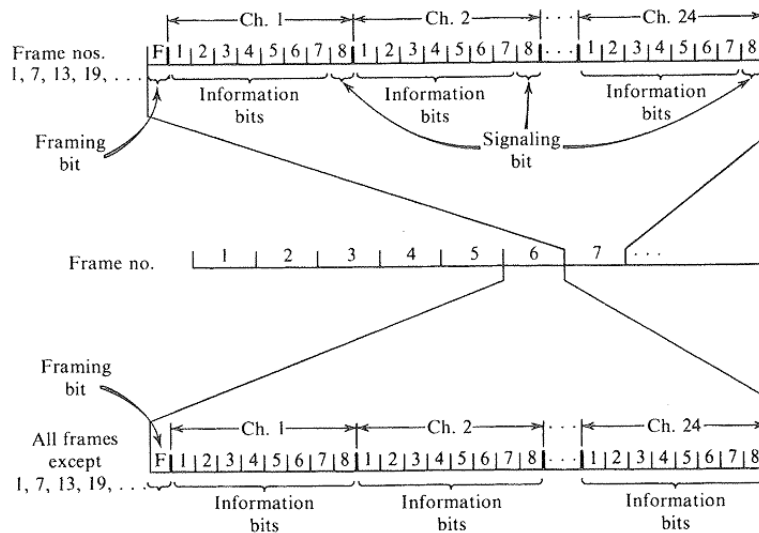


Figure 6.26. T1 system signaling format.

6.9.3. Digital Hierarchy

The following is the digital hierarchy developed by the Bell System and used in North America and Japan. 4 DS1 signals are multiplexed to yield a DS2 signal at a rate of 6.312 Mb/s. 7 DS2 signals are multiplexed to yield a DS3 signal at 44.736 Mb/s. Finally, three DS3 signals are multiplexed to yield a DS4 signal at 139.264 Mb/s. Note that the inputs to the multiplexer that gives DS2 signal need not be digital voice channels: Any digital signal of 64 kb/s of appropriate format can be transmitted. The same goes for higher rate signal: The input to give DS2 signals need not be DS1 signals: Anything with 1.544 Mb/s with the appropriate format will be ok. This is essentially how we use our

Telephone to connect to the Internet back in the 1990's (remember we have a box called modem that needs to connect to the telephone jack and it will make some strange sounds before the computer got connected to the Internet?)

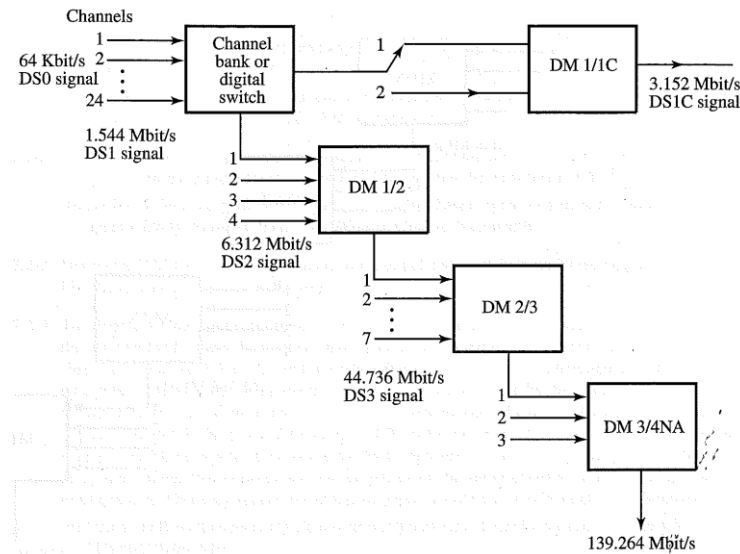


Figure 6.27. North American digital hierarchy (AT&T system)

Example 6.2

For the system as shown in the following figure which is a complete PCM system with 10 time-multiplexed signal channels, find the minimum sampling rate. Assuming that 30,000 samples / s are taken, find the bit rate at the PCM output.

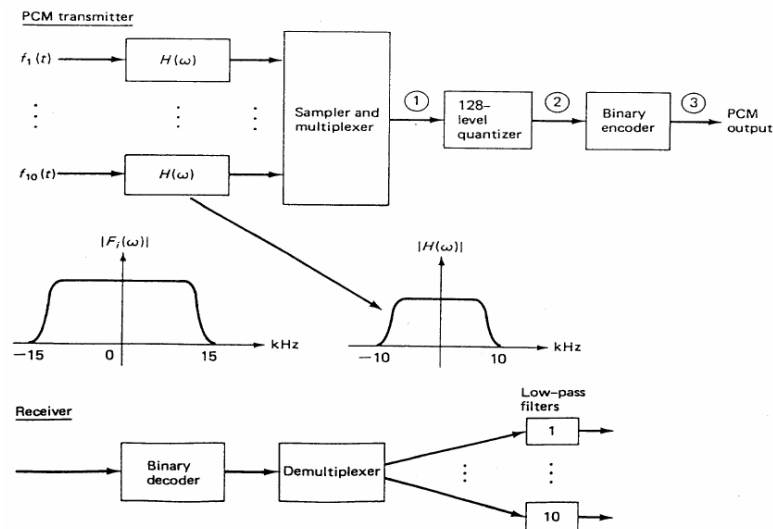


Figure 6.28. Figure for Example 5.3

Chapter 7 Digital Communication

Objectives

- To describe a binary digital transmission system
- To explain intersymbol interference
- To show how an eye diagram is produced and explain its significance for system performance assessment
- To introduce the condition for zero intersymbol interference
- To introduce the term "error probability"
- To introduce the concept of error control coding
- To introduce the concept of line coding and to provide illustrations of some practical line codes

References: Chapter 7,8 of S. Haykin and M. Moher, *Communication Systems*, John Wiley and Sons, 5th edition.

The pulse modulation systems such as PAM, PWM and PPM are not completely digital, as the amplitude, width, or position of the pulses transmitted may vary continuously in accordance with a message signal variation. PAM is the simplest pulse modulation system. For this reason, the PAM signals are used to get a complete digital system. The PAM signals are first quantized and then coded (usually in binary code), thus, giving rise to PCM system. The output of a PCM system is a series of binary digits, i.e., a string of 1's and 0's. These binary digits are transmitted over the transmission channel.

7.1 Binary digital transmission system

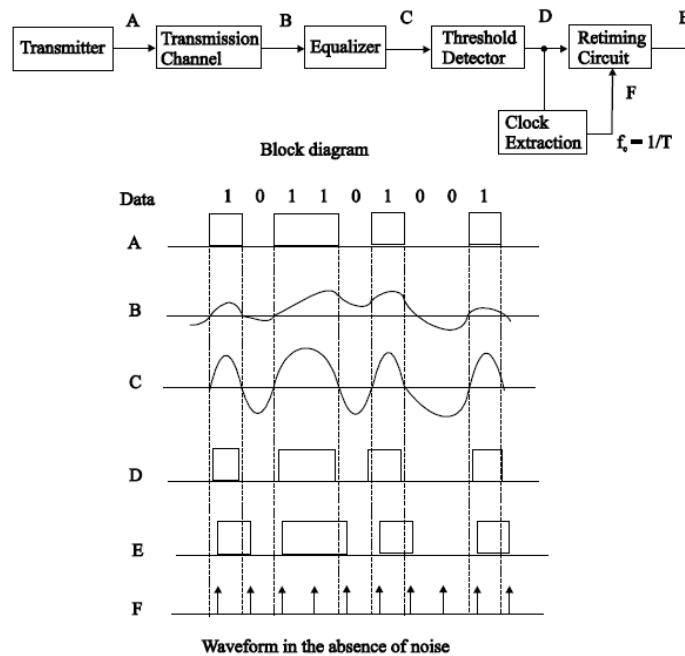


Figure 7.1 Signal waveforms at different points in a digital communication system.

The transmitted signal, when it arrived at the output of the channel, is attenuated and distorted due to the loss and restricted bandwidth of the channel. It is the role of the receiver to recover the original data.

The distortion can be partially corrected by the use of an *equalizer*. This equalized waveform is then applied to a threshold device which restores the signal to its original form except that the timing of the transitions may be irregular. This arises because the transition times depend on the transmission channel and the threshold circuit which may be corrupted by the noise. Hence, to restore the relative regularity of data transitions a clock waveform of frequency $f_c = 1/T$ is extracted from the signal and used to retime the data. This combination of threshold and retiming circuitry provides for data recovery and corresponds to making decisions on the basis of samples of the signal taken at regularly

spaced interval of T seconds, the sampling epochs being adjusted to provide data extraction at the peaks and troughs of the equalized signal waveform.

7.2 Source coding for digital transmission

In binary PCM, each sample of the signal is represented by a codeword of, say, k bits. These bits are transmitted and the role of the receiver is to recognize each codeword in order to reconstruct the samples.

Source coding is the process to transform/represent a source of information into bits.

7.2.1. Digital representation of text

Encoding English text use Seven bit ASCII (American Standard Code for Information Interchange) code.

ASCII Code: Character to Binary

0	0011 0000	O	0100 1111	m	0110 1101
1	0011 0001	P	0101 0000	n	0110 1110
2	0011 0010	Q	0101 0001	o	0110 1111
3	0011 0011	R	0101 0010	p	0111 0000
4	0011 0100	S	0101 0011	q	0111 0001
5	0011 0101	T	0101 0100	r	0111 0010
6	0011 0110	U	0101 0101	s	0111 0011
7	0011 0111	V	0101 0110	t	0111 0100
8	0011 1000	W	0101 0111	u	0111 0101
9	0011 1001	X	0101 1000	v	0111 0110
A	0100 0001	Y	0101 1001	w	0111 0111
B	0100 0010	Z	0101 1010	x	0111 1000
C	0100 0011	a	0110 0001	y	0111 1001
D	0100 0100	b	0110 0010	z	0111 1010
E	0100 0101	c	0110 0011	.	0010 1110
F	0100 0110	d	0110 0100	,	0010 0111
G	0100 0111	e	0110 0101	:	0011 1010
H	0100 1000	f	0110 0110	;	0011 1011
I	0100 1001	g	0110 0111	?	0011 1111
J	0100 1010	h	0110 1000	!	0010 0001
K	0100 1011	I	0110 1001	'	0010 1100
L	0100 1100	j	0110 1010	"	0010 0010
M	0100 1101	k	0110 1011	{	0010 1000
N	0100 1110	l	0110 1100	}	0010 1001
				space	0010 0000

Figure 7.2. Signal waveforms at different points in a digital communication system.

7.2.2. Digital representation of images – pixels and image resolutions

In digital imaging, a pixel, (picture element) is a single point in an image, or the smallest addressable screen element in a display device; it is the smallest unit of picture that can be represented or controlled.

Each pixel has its own address. The address of a pixel corresponds to its coordinates. Pixels are normally arranged in a two-dimensional grid, and are often represented using dots or squares. Each pixel is a sample of an original image; more samples typically provide more accurate representations of the original. The intensity of each pixel is variable. In color image systems, a color is typically represented by three or four component intensities such as red, green, and blue, or cyan, magenta, yellow, and black. In some contexts (such as descriptions of camera sensors), the term pixel is used to refer to a single scalar element of a multi-component representation while in others the term may refer to the entire set of such component intensities for a spatial position.

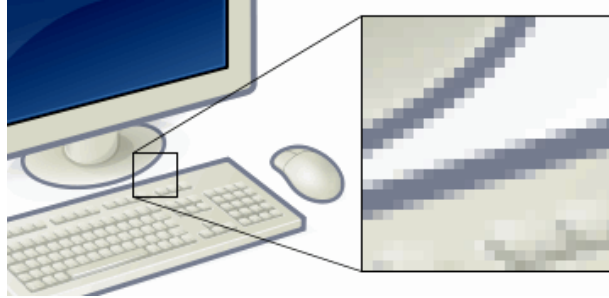


Figure 7.3. Each digital image consist of a large number of pixels.

The number of distinct colors that can be represented by a pixel depends on the number of bits per pixel (bpp). A 1 bpp image uses 1-bit for each pixel, so each pixel can be either on or off. Each additional bit doubles the number of colors available, so a 2 bpp image can have 4 colors, and a 3 bpp image can have 8 colors:

- 1 bpp, $2^1 = 2$ colors (monochrome)
- 2 bpp, $2^2 = 4$ colors
- 3 bpp, $2^3 = 8$ colors
- 8 bpp, $2^8 = 256$ colors
- 16 bpp, $2^{16} = 65,536$ colors ("Highcolor")
- 24 bpp, $2^{24} \approx 16.8$ million colors ("Truecolor")

The display resolution of a digital television or display device is the number of distinct pixels in each dimension that can be displayed. It can be an ambiguous term especially as the displayed resolution is controlled by all different factors in cathode ray tube (CRT), flat panel or projection displays using fixed picture-element (pixel) arrays.

It is usually quoted as width $\times$ height, with the units in pixels: for example, "1024x768" means the width is 1024 pixels and the height is 768 pixels. This example would normally be spoken as "ten twenty-four by seven sixty-eight" or "ten twenty-four by seven six eight".

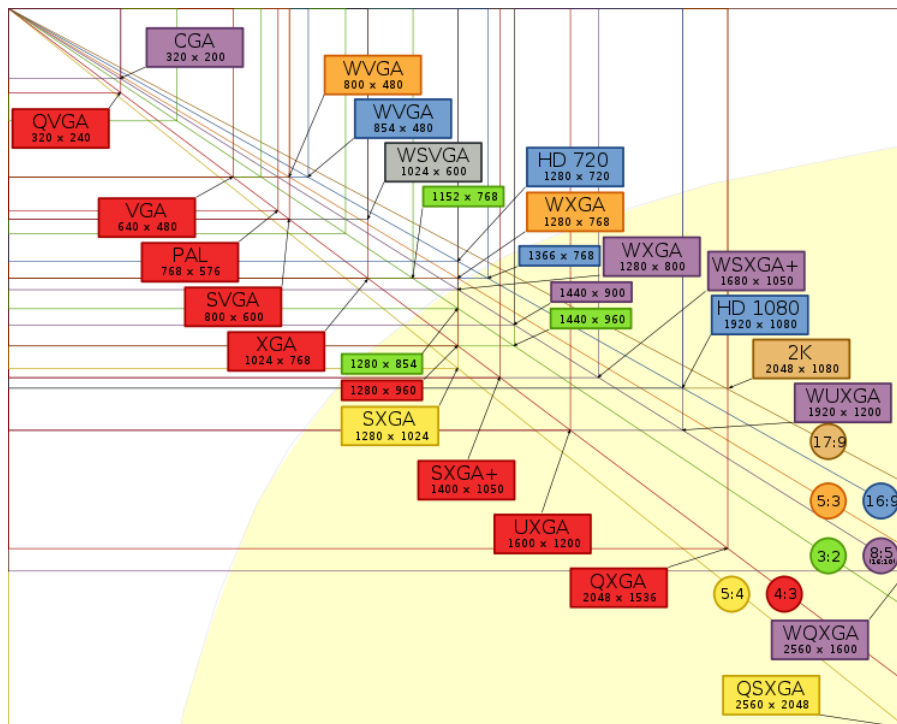


Figure 7.4. Resolution of different video formats (in terms of number of pixels x number of pixels).

7.3 Practical Pulse Shaping

The digital data need to be coded into electrical pulses or waveforms for the purpose of transmission over the channel. This process is called **line coding** or **transmission coding**. This type of coding is used to match the digital signal to the physical characteristics of the channel (a communication link usually cannot transmit dc components) and facilitate synchronization at the receiver. The receiver must know the rate at which data bits arrive (known as the “clock rate”) so that it can achieve synchronization.

7.3.1. Unipolar code

The simplest line code is on-off, where a “1” is transmitted by a pulse and a “0” is transmitted by no pulse.

7.3.2. Polar code

“1” is transmitted by a pulse $p(t)$ and “0” is transmitted by a pulse $-p(t)$.

7.3.3. Bipolar code

“0” is encoded by no pulse and “1” is encoded by a pulse $p(t)$ or $-p(t)$, depending on whether the previous “1” is encoded by $p(t)$ or $-p(t)$. In short, pulse representing consecutive 1’s alternate in sign.

The *non-return-to-zero (NRZ)* format has disadvantages because it has a dc component.

The *NRZ-bipolar (NRZ-B)* format eliminates the dc component (assuming that an equal number of 1s and 0s occur)

The *return-to-zero (RZ)* format includes clock information, has zero dc level (again, for equal numbers of 1s and 0s), but has three voltage levels (for bipolar) and requires twice the bandwidth of the NRZ formats.

Now, how do we mathematically express these waveforms? what is the bandwidth of these signals? For a given pulse shape $p(t)$ with pulse duration T , the overall waveform can be written as

$$s(t) = \sum_{n=-\infty}^{\infty} x_n p(t - nT)$$

Where x_n are the binary data sequence ‘01100111010100101.....’

So, what is the bandwidth of $s(t)$?

$$S(\omega) = \sum_{n=-\infty}^{\infty} x_n P(\omega) e^{j\omega nT} = P(\omega) \cdot \sum_{n=-\infty}^{\infty} x_n e^{j\omega nT}$$

In fact, with some more algebra which is too tedious to show here, one can simply say the bandwidth of $s(t)$ is equal to the bandwidth $p(t)$.

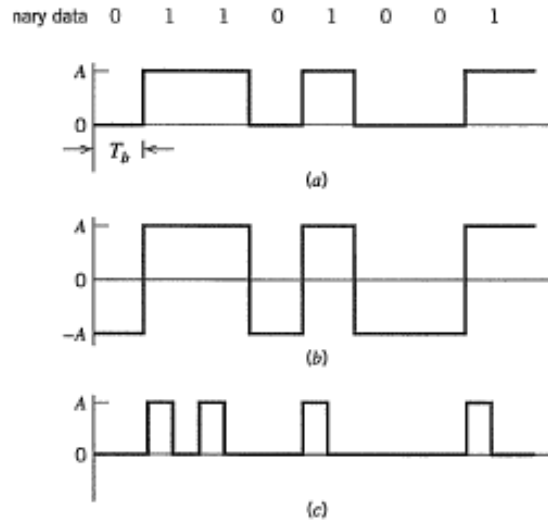


Figure 7.5. (a) Unipolar NRZ pulses (b) Polar NRZ pulse (c) Unipolar RZ pulse.

7.4 Inter-symbol interference (ISI)

When a signal is transmitted over a channel of limited bandwidth, a significant portion of its spectrum is transmitted, but a small portion is suppressed. Such a spectral distortion tends to spread the pulse. Spreading of a pulse beyond its interval causes it to interfere with neighboring pulses. This is known as **inter-symbol interference (ISI)**.

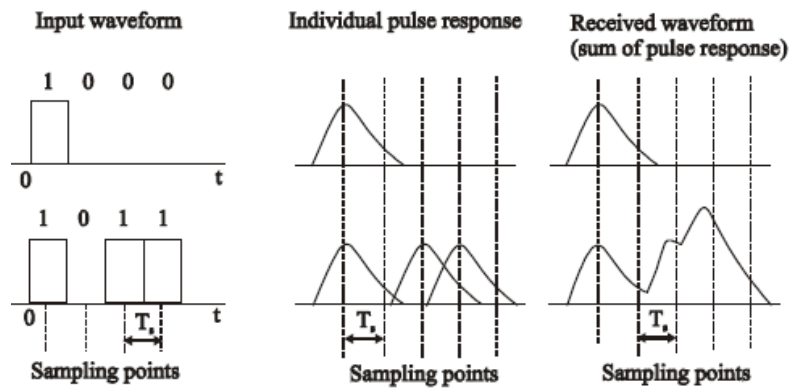


Figure 7.6. Inter-symbol interference (ISI) of individual pulse will affect detection of binary signals.

Both the inter-symbol interference and noise (which will be discussed later) may cause errors, and hence affect the system performance.

7.4.1. Condition for zero intersymbol interference

If the pulse shape is $h(t)$ with pulse period T_b , then $h(t)$ must satisfy

$$h(0) = \text{non-zero constant}$$

$$h(nT_b) = 0, \quad \text{when } n \neq 0.$$

Pulse satisfying the condition for zero ISI:

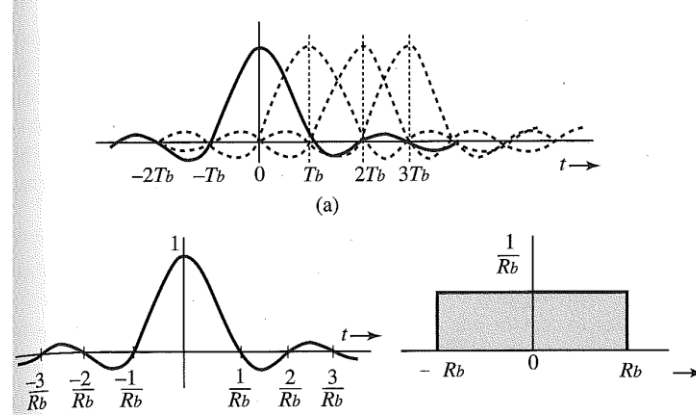


Figure 7.7. Pulse shape with zero inter-symbol interference. Among all the pulse with zero ISI, the sinc pulse has the minimum bandwidth.

It is theoretically possible to obtain both zero ISI and limited bandwidth usage. The sinc pulse has the smallest bandwidth for all zero-ISI pulses. In practice however, it is not possible to obtain this pulse shape because it is the impulse response of an ideal low-pass filter.

7.5 Raised Cosine pulses

Practical ideal Nyquist channel difficulties can be eliminated by increasing the minimum bandwidth from π / T rad/s or $W = 1/2 T$ Hz to an adjustable value between W and $2W$. A particular spectrum $P(f)$ of a pulse $p(t)$ can be used to decrease ISI, has a few conditions to be satisfied, and is known as a **raised cosine spectrum**. $P(f)$ has a few portions, a flat portion and a sinusoidal roll-off portion, specified by:

$$P(f) = \begin{cases} \frac{1}{2W}, & 0 \leq |f| < f_1 \\ \frac{1}{4W} \left\{ 1 - \sin \left[\frac{\pi(|f| - W)}{2W - 2f_1} \right] \right\}, & f_1 \leq |f| < 2W - f_1 \\ 0, & |f| \geq 2W - f_1 \end{cases}$$

The frequency parameter f_1 and bandwidth W are related by

$$\alpha = 1 - \frac{f_1}{W}$$

The parameter α is known as the roll-off factor, which shows how much more bandwidth there is over the ideal solution W . The transmission bandwidth B_T is defined by $2W - f_1 = W(1 + \alpha)$.

Figure 7.8 (a) shows the normalized frequency response $P(f)$ for three α values: 0, 0.5, and 1. $P(f)$ is easy to implement in practice for α values for 0.5 and 1 because $P(f)$ would be cut off gradually (Nyquist channel has $\alpha = 0$).

The time response $p(t)$ is given by:

$$p(t) = [\text{sinc}(2Wt)] \left(\frac{\cos(2\pi\alpha Wt)}{1 - 16\alpha^2 W^2 t^2} \right)$$

and is plotted in Figure 7.8(b) for $\alpha = 0, 0.5$, and $|t| < 3T$.

$p(t)$ is the product of two factors: $\text{sinc}(2Wt)$ which characterizes the ideal Nyquist channel, and the second factor which decreases for large $|t|$ due to t^2 being in the denominator. The first factor ensures zero crossings of $p(t)$ at the desired sampling instants of time $t = kT$, where k is an arbitrary integer. The second factor reduce pulse tails much more lower than the ideal Nyquist channel so that binary wave transmission using these pulses is not as sensitive to sampling time errors. For $\alpha = 1$, we have the most gradual roll-off since the oscillatory tails of $p(t)$ are the smallest. As a result, ISI due to timing error decreases because the roll-off factor α is increased from zero to one.

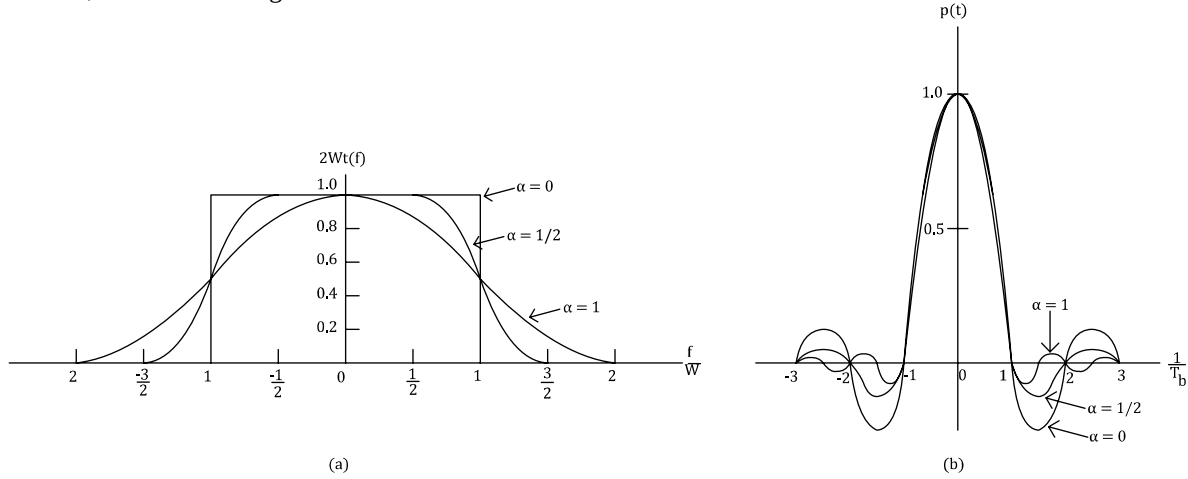


Figure 7.8. Pulses corresponding to different roll-off factors. (a) Frequency response. (b) Impulse response.

Example 7.1. Bandwidth Requirement of the T1 System

In the previous chapter, we described the signal format for the T1 carrier system that is used to multiplex 24 independent voice inputs, based on an 8-bit PCM byte. It was shown that the bit duration of the resulting time-division multiplexed signal (including a framing bit) is

$$T_b = 0.647 \mu\text{s}$$

Assuming the use of an ideal Nyquist channel, it follows that the minimum transmission bandwidth B_T of the T1 system is (for $\alpha = 0$)

$$B_T = W = \frac{1}{2T_b} = 772 \text{ kHz}$$

However, a more realistic value for the necessary transmission bandwidth is obtained by using a full-cosine rolloff characteristic with $\alpha = 1$. In this case, we find that

$$B_T = W(1 + \alpha) = 2W = \frac{1}{T_b} = 1.544 \text{ MHz}$$

It is interesting to compare the transmission bandwidth requirement of the T1 system with the minimum bandwidth requirement of a corresponding frequency-division multiplexing (FDM) system. We recall from Chapter 3 that of all the CW modulation techniques, the use of single side-band (SSB) modulation requires the minimum bandwidth possible. Thus, to accommodate an FDM system using SSB modulation to transmit 24 independent voice inputs, and assuming a bandwidth of 4 kHz for each voice input, the channel must provide the transmission bandwidth

$$B_T = 24 \times 4 = 96 \text{ kHz}$$

This is more than an order of magnitude smaller than the bandwidth requirement of the T1 system.

7.6 The eye diagram

The *eye diagram* is the synchronized superposition of all possible signal patterns obtained in a particular signal interval. The eye pattern derives its name from the fact that it resembles the human eye for binary waves.

The interior region of the eye pattern is called the *eye opening*. Eye opening is also the *timing error allowed on the sampler at the receiver*. Under normal operating conditions (i.e., for no detection errors), the eye pattern will *open*. If

there is great deal of noise or ISI, the eye will *close*; this indicates that bit errors will be produced at the receiver output.

An eye pattern provides a great deal of useful information about the performance of a digital transmission system.

- The width of the eye opening defines the time interval over which the received signal can be sampled without error from ISI. The best sampling time is the instant where the vertical opening of the eye is the largest.
- The height of the eye opening, at a specified sampling time, defines the *noise margin* of the system. If noise is larger than the noise margin, there will be bit error
- The slope of the eye pattern indicates the sensitivity to timing error.

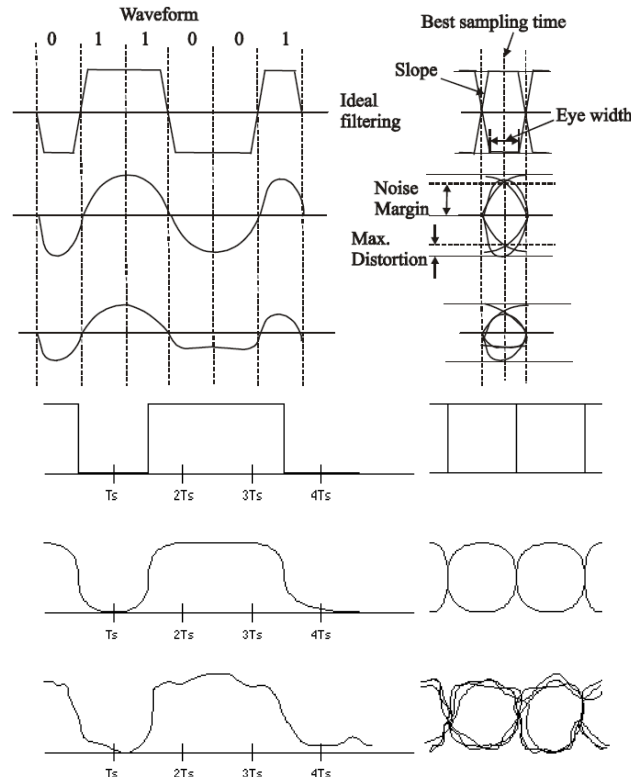
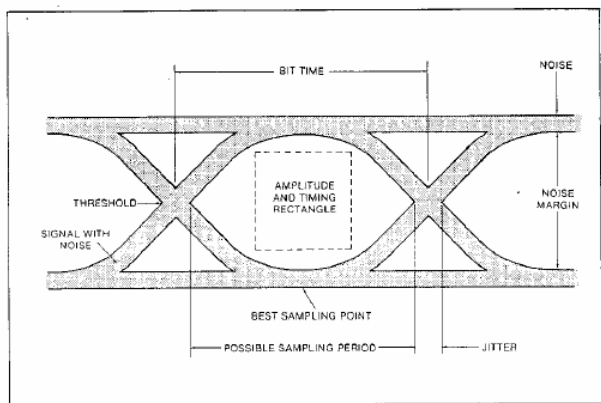
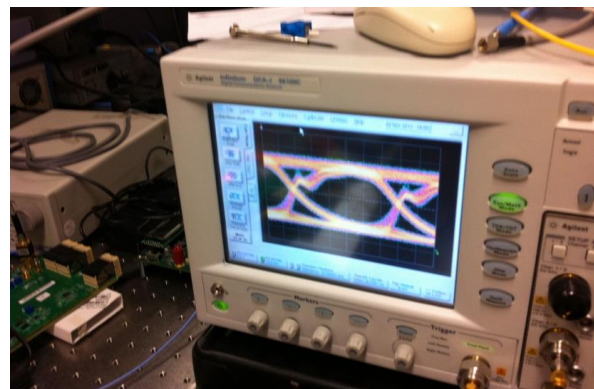


Figure 7.9. Distorted waveforms and corresponding eye diagrams.



(a)



(b)

Figure 7.10. (a) Different aspects and features of the eye diagram; (b) Eye diagrams obtained in actual experiments as a way to measure system performance;